

QUESTION 6(a)

Question:

To Create Process Escape Sequences, Handle Escaped Spaces, Escape Special Symbols, Preserve Characters, Validate Parser Output, Test Complex Inputs.

```
#include <stdio.h>
#include <string.h>

#define MAX 200

int main()
{
    char input[MAX];
    char output[MAX];

    int i = 0;
    int j = 0;

    printf("==== Escape Sequence Parser ====\\n");
    printf("Enter text with escape characters: ");

    fgets(input, MAX, stdin);
    input[strcspn(input, "\\n")] = '\\0';

    while (input[i] != '\\0' && j < MAX - 1)
    {
        if (input[i] == '\\\\')
        {
            i++;

            if (input[i] == '\\0')
            {
                printf("\\nError: Escape character at end.\\n");
                return 1;
            }
        }
    }
}
```

```

/* Handle escaped space */
if (input[i] == ' ')
{
    output[j++] = ' ';
}

/* Handle escaped special symbols */
else if (input[i] == '\\' ||
         input[i] == '|' ||
         input[i] == '&' ||
         input[i] == ';' ||
         input[i] == '"' ||
         input[i] == '\\')
{
    output[j++] = input[i];
}

/* Handle newline */
else if (input[i] == '\n')
{
    output[j++] = '\n';
}

/* Handle tab */
else if (input[i] == '\t')
{
    output[j++] = '\t';
}

/* Unknown escape */
else
{
    printf("\nWarning: Unknown escape \\%c\n",
           input[i]);

    output[j++] = input[i];
}
}

```

```

    }
    else
    {
        output[j++] = input[i];
    }

    i++;
}

output[j] = '\0';

printf("\nOriginal Input:\n%s\n", input);

printf("\nParsed Output:\n%s\n", output);

printf("\nEscape sequence processing completed.\n");

return 0;
}

```

Output

```

sri_dhruthi@Dhruthi:~$ nano skill6a.c
sri_dhruthi@Dhruthi:~$ gcc skill6a.c -o skill6a
sri_dhruthi@Dhruthi:~$ ./skill6a
===== Escape Sequence Parser =====
Enter text with escape characters: Hello\ World

Original Input:
Hello\ World

Parsed Output:
Hello World

Escape sequence processing completed.
sri_dhruthi@Dhruthi:~$ ./skill6a
===== Escape Sequence Parser =====
Enter text with escape characters: Hello\|World

Original Input:
Hello\|World

Parsed Output:
Hello|World

Escape sequence processing completed.

```

QUESTION 6(b)

Question:

To Create Child Processes, Execute Programs, Handle Execution Errors, Pass Arguments, Manage Parent Process, Test Command Launching.

```
GNU nano 2.7.1 SRICOD.C
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    pid_t pid;
    int status;

    printf("==== Process Execution ====\\n");

    pid = fork();

    if (pid < 0)
    {
        perror("fork failed");
        return 1;
    }

    /* Child process */
    if (pid == 0)
    {
        printf("\\nChild Process Created\\n");
        printf("Child PID: %d\\n", getpid());
        printf("Parent PID: %d\\n", getppid());

        printf("\\nExecuting ls command...\\n");

        execlp("ls", "ls", "-l", NULL);

        /* Executes only if execlp fails */
        perror("Execution failed");

        exit(1);
    }
}
```

```

        exit(1);
    }

    /* Parent process */
    else
    {
        printf("\nParent Process\n");
        printf("Parent PID: %d\n", getpid());
        printf("Child PID: %d\n", pid);

        printf("\nWaiting for child process...\n");

        waitpid(pid, &status, 0);

        if (WIFEXITED(status))
        {
            printf("\nChild exited with status: %d\n",
                WEXITSTATUS(status));
        }
        else
        {
            printf("\nChild terminated abnormally.\n");
        }

        printf("Parent process completed.\n");
    }

    return 0;
}

```

```

sri_dhruthi@Dhruthi:~$ nano skill6b.c
sri_dhruthi@Dhruthi:~$ gcc skill6b.c -o skill6b
sri_dhruthi@Dhruthi:~$ ./skill6b
===== Process Execution =====

```

```

Parent Process
Parent PID: 1116
Child PID: 1117

```

```

Waiting for child process...

```

```

Child Process Created
Child PID: 1117
Parent PID: 1116

```

```

Executing ls command...

```

```

total 828
-rwxr-xr-x 1 sri_dhruthi sri_dhruthi 16000 Aug 16 10:54 ExecExample
-rw-r--r-- 1 sri_dhruthi sri_dhruthi   181 Aug 16 11:00 ExecExample.c
-rwxr-xr-x 1 sri_dhruthi sri_dhruthi 16056 Aug 16 10:52 MultipleChildProcess
-rw-r--r-- 1 sri_dhruthi sri_dhruthi   238 Aug 16 11:00 MultipleChildProcess
.c
-rwxr-xr-x 1 sri_dhruthi sri_dhruthi 16056 Aug 14 10:08 Multiplechildprocess
-rw-r--r-- 1 sri_dhruthi sri_dhruthi   258 Aug 14 10:06 Multiplechildprocess

```

```

-rwxr-xr-x 1 sri_dhruthi sri_dhruthi 16056 Aug 14 07:12 wait_example.c
-rw-r--r-- 1 sri_dhruthi sri_dhruthi   282 Aug 14 07:24 wait_example.c

```

```

Child exited with status: 0
Parent process completed.

```